

**ATTACHMENT 33 – LETTER FROM GANNETT FLEMING, INC. TO BECHTEL
/ PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) DATED
AUGUST 25, 1997 REGARDING PROPOSAL FOR ALTERNATE ANCHOR
SYSTEM
(4 pages)**



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August 25, 1997

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Mr. John Bartoletti
Authorized Representative
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One South Station
Boston, MA 02110

Subject: Central Artery (I-93/Tunnel I-90) Project
Fort Point Channel Crossing
Project No. D009B
9B2 Post-Installed Anchoring System for Hanging Ceiling in the 4A2 Tunnel

Dear Mr. Bartoletti:

Gannett Fleming has been researching various types of post-installed anchoring systems for the subject tunnel. Available resources include manufacturers catalogs, proprietary test data, manufacturer's engineering departments and ACI publications. We focused on three types of anchors, the resin bonded anchor, the undercut anchor and the expansion anchor. The selected anchor will be installed in the tensile zone of the concrete roof slab. The anchors may also fall in cracks which may form in the concrete. Project Design Criteria also requires that materials located in the exhaust plenum be capable of withstanding 600°F for 1 hour. Also, according to Project Design Criteria the maximum crack width expected in the underside of this roof slab is approximately .40 mm.

Summarized below are Gannett Fleming's findings for each anchor system focusing on the anchors being loaded in tension only. Actual anchor design will include combined loading cases.

RESIN BONDED ANCHORS

Resin bonded anchor systems consist of a two component adhesive capsule and an anchor rod assembly with nut & washer. The tensile load is transferred from the anchor to the concrete substrate by the hardened resin mortar.

A research report which we reviewed stated that a common resin bonded anchor under axial tension in cracked concrete will only retain 40% of the anchors ultimate load capacity in uncracked concrete and may retain as little as 20%. Manufacturers literature noted that their

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resins lose bond strength when the concrete temperature is between 120°F and 230°F. Meeting high temperature requirements as well as maintaining design capacity appears difficult with resin bonded anchors. Gannett Fleming will continue to investigate the availability of other alternatives, such as high heat resins.

UNDERCUT ANCHORS

Undercut anchors consist of a threaded rod with nut assembly, expansion sleeve and expansion plug. When the nut is tightened, the expansion plug expands the steel expansion sleeve into the undercut at the top of the drilled hole. Tensile forces are transferred into the concrete by keying and bearing against the undercut in the concrete.

A research report which we reviewed stated that an undercut anchor in cracked concrete will retain 65% to 70% of the anchors ultimate capacity in uncracked concrete. Undercut anchors perform the same as cast-in headed anchors according to this paper. These anchors will perform similar to steel when exposed to high temperatures.

Undercut anchors are recommended by manufacturers for application in the tensile zone of concrete as well as cracked concrete. They are recommended for dynamic loading as well. These anchors are available in a variety of steel materials and with zinc plating or hot-dipped galvanized coatings depending on the manufacturer. These anchors appear to have the desired characteristics for supporting the ceiling system in the subject tunnel.

EXPANSION TYPE ANCHORS

Expansion type anchors are available in a variety of styles. With these anchors the tensile load is transferred from the anchor to the concrete substrate by friction.

A research report which we reviewed stated that torque controlled expansion anchors under tension with no capacity to post-expand in cracked concrete are unsuitable for use. In this situation neither the ultimate load nor the load displacement behavior of the anchor can be forecast accurately. We are unaware of any torque controlled anchors available through U.S. Manufacturers which have post-expansion capabilities.

We found one foreign manufacturer who claims to have a torque controlled expansion anchor which is suitable for application in cracks. With a crack width of 0.40 mm the ultimate load is reduced to 60% of the value in uncracked concrete. With increased crack widths the reduction in ultimate load capacity continues to occur.

Gannett Fleming

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We would like to meet with your structural engineers as soon as possible to discuss our findings and to obtain direction on the anchor preferred by B/PB.

Very truly yours,



Donald B. Nicholas, P.E.
Project Manager

EVD:asb

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cc: P. J. Falk - GF
P. D. Smith - GF
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